

Tutorial 4

Determinants

Homework

1. Provide detailed solutions to *two* selected parts from Q1 (parts 6–15).
2. Provide detailed solutions to Q4 and Q5.
3. Provide detailed solutions to *two* of the following questions: Q6–Q9.
4. Provide detailed solutions to *two* of the following questions: Q11–Q20.
5. Provide detailed solutions to Q23.
6. Provide a detailed solution to *one* selected part of Q25.
7. Provide a detailed solution to *one* selected part of Q28.
8. Provide detailed solutions to *three* selected parts from Q33.
9. Provide detailed solutions to Q34, part 4.
10. Provide detailed solutions to *three* selected parts from Q35.

Question 1. Compute the determinants of the following matrices using the standard determinant formula for 2×2 matrices and Sarrus' rule for 3×3 matrices.

- | | | |
|--|---|--|
| 1. $\begin{pmatrix} 2 & 5 \\ 3 & 1 \end{pmatrix}$ | 2. $\begin{pmatrix} 5 & -3 \\ 2 & 0 \end{pmatrix}$ | 3. $\begin{pmatrix} 6 & -12 \\ -4 & 8 \end{pmatrix}$ |
| 4. $\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ | 5. $\begin{pmatrix} 2 & 0 & 5 \\ -4 & 1 & 7 \\ 0 & 3 & -3 \end{pmatrix}$ | 6. $\begin{pmatrix} 3 & -2 & 4 \\ 5 & 1 & -2 \\ -1 & 3 & 6 \end{pmatrix}$ |
| 7. $\begin{pmatrix} 5 & 0 & 0 \\ 3 & -2 & 0 \\ -1 & 8 & 4 \end{pmatrix}$ | 8. $\begin{pmatrix} -6 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ | 9. $\begin{pmatrix} 3 & 1 & -2 \\ -1 & 4 & 5 \\ 3 & 1 & -2 \end{pmatrix}$ |
| 10. $\begin{pmatrix} \alpha - 3 & 5 \\ -3 & \alpha - 2 \end{pmatrix}$ | 11. $\begin{pmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{pmatrix}$ | 12. $\begin{pmatrix} -2 & 7 & 6 \\ 5 & 1 & -2 \\ 3 & 8 & 4 \end{pmatrix}$ |
| 13. $\begin{pmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{pmatrix}$ | 14. $\begin{pmatrix} -2 & 1 & 4 \\ 3 & 5 & -7 \\ 1 & 6 & 2 \end{pmatrix}$ | 15. $\begin{pmatrix} \beta & -4 & 3 \\ 2 & 1 & \beta^2 \\ 4 & \beta - 1 & 2 \end{pmatrix}$ |

Solution.

1. $\begin{vmatrix} 2 & 5 \\ 3 & 1 \end{vmatrix} = 2 \times 1 - 3 \times 5 = -13$
2. 6
3. 0
4. $\cos^2 \theta + \sin^2 \theta = 1$
5. $2 \times 1 \times (-3) + 0 + 5 \times (-4) \times 3 - 0 - 2 \times 3 \times 7 - 0 = -6 - 60 - 42 = -108$

- | | | | |
|-------------------------------|-----------|---|------|
| 6. 156 | 7. -40 | 8. -60 | 9. 0 |
| 10. $\alpha^2 - 5\alpha + 21$ | 11. -4 | 12. 0 | |
| 13. -123 | 14. -65 | 15. $-\beta^4 + \beta^3 - 16\beta^2 + 8\beta - 2$ | |

Question 2. Find all the minors and cofactors of the matrix A .

$$1. A = \begin{pmatrix} 1 & -2 & 3 \\ 6 & 7 & -1 \\ -3 & 1 & 4 \end{pmatrix}, \quad 2. A = \begin{pmatrix} 1 & 1 & 2 \\ 3 & 3 & 6 \\ 0 & 1 & 4 \end{pmatrix}$$

Solution.

1. Minors

$$M_{11} = \begin{vmatrix} 7 & -1 \\ 1 & 4 \end{vmatrix} = 29, \quad M_{12} = 21, \quad M_{13} = 27, \quad M_{21} = -11, \quad M_{22} = 13,$$

$$M_{23} = -5, \quad M_{31} = -19, \quad M_{32} = -19, \quad M_{33} = 19.$$

Cofactors:

$$C_{11} = (-1)^{1+1} \times 29 = 29, \quad C_{12} = -21, \quad C_{13} = 27, \quad C_{21} = 11, \quad C_{22} = 13, \quad C_{23} = 5,$$

$$C_{31} = -19, \quad C_{32} = 19, \quad C_{33} = 19$$

2. Minors

$$M_{11} = 6, \quad M_{12} = 12, \quad M_{13} = 3, \quad M_{21} = 2, \quad M_{22} = 4,$$

$$M_{23} = 1, \quad M_{31} = 0, \quad M_{32} = 0, \quad M_{33} = 0.$$

Cofactors:

$$C_{11} = 6, \quad C_{12} = -12, \quad C_{13} = 3, \quad C_{21} = -2, \quad C_{22} = 4, \quad C_{23} = -1,$$

$$C_{31} = 0, \quad C_{32} = 0, \quad C_{33} = 0$$

Question 3. Let

$$A = \begin{pmatrix} 2 & 3 & -1 & 1 \\ -3 & 2 & 0 & 3 \\ 3 & -2 & 1 & 0 \\ 3 & -1 & 1 & 4 \end{pmatrix}$$

Find

1. Minor M_{32} and cofactor C_{32}
2. Minor M_{41} and cofactor C_{41}
3. Minor M_{44} and cofactor C_{44}
4. Minor M_{24} and cofactor C_{24}

Solution.

1.

$$M_{32} = \begin{vmatrix} 2 & -1 & 1 \\ -3 & 0 & 3 \\ 3 & 1 & 4 \end{vmatrix} = 3 \times \begin{vmatrix} -1 & 1 \\ 1 & 4 \end{vmatrix} - 3 \times \begin{vmatrix} 2 & -1 \\ 3 & 1 \end{vmatrix} = 3 \times (-5) - 3 \times 5 = -30, \quad C_{32} = (-1)^{3+2} M_{32} = 30$$

2. $M_{41} = -1, C_{41} = 1$

3. $M_{44} = 13, C_{44} = 13$

4. $M_{24} = -5, C_{24} = -5$

Question 4. For each of the given matrices A, B, C , find the specified minors and cofactors.

$$A = \begin{pmatrix} 2 & 4 & 3 \\ 3 & -1 & 6 \\ 5 & -2 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 2 & -3 & 1 \\ 1 & 4 & 2 & -1 \\ 3 & -2 & 4 & 0 \\ 4 & -1 & 1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} -3 & 3 & 0 & 5 \\ 2 & 1 & -1 & 4 \\ 6 & -3 & 4 & 0 \\ -1 & 5 & 1 & -2 \end{pmatrix}$$

1. Minors $M_{12}, M_{21}, M_{23}, M_{31}$ and cofactors $C_{12}, C_{22}, C_{32}, C_{21}$ for matrix A .2. Minors $M_{12}, M_{42}, M_{24}, M_{31}, M_{34}$ and cofactors $C_{12}, C_{24}, C_{31}, C_{42}, C_{33}$ for matrix B .3. Minors $M_{11}, M_{41}, M_{23}, M_{32}, M_{44}$ and cofactors $C_{12}, C_{24}, C_{31}, C_{42}, C_{33}$ for matrix C .

Solution.

1.

$$M_{12} = -18, \quad M_{21} = 22, \quad M_{23} = -24, \quad M_{31} = 27,$$

$$C_{12} = 18, \quad C_{22} = -7, \quad C_{32} = -3, \quad C_{21} = -22$$

2.

$$M_{12} = 13, \quad M_{42} = 7, \quad M_{24} = 11, \quad M_{31} = 5, \quad M_{34} = 65$$

$$C_{12} = -13, \quad C_{24} = 11, \quad C_{31} = 5, \quad C_{42} = 7, \quad C_{33} = -25$$

3.

$$M_{11} = -94, \quad M_{41} = -43, \quad M_{23} = 153, \quad M_{32} = 11, \quad M_{44} = -45$$

$$C_{12} = -12, \quad C_{24} = 39, \quad C_{31} = 24, \quad C_{42} = 118, \quad C_{33} = 121$$

Question 5. Find the minors M_{11}, M_{31}, M_{23} and cofactors C_{12}, C_{32}, C_{22} for the following matrix

$$A = \begin{pmatrix} \alpha + 1 & \alpha & \alpha - 7 \\ \alpha - 4 & \alpha + 5 & \alpha - 3 \\ \alpha - 1 & \alpha & \alpha + 2 \end{pmatrix}$$

Solution.

$$M_{11} = \begin{vmatrix} \alpha + 5 & \alpha - 3 \\ \alpha & \alpha + 2 \end{vmatrix} = (\alpha + 2)(\alpha + 5) - \alpha(\alpha - 3) = 7\alpha + 10 + 3\alpha = 10\alpha + 10, \quad M_{31} = 35 - \alpha, \quad M_{23} = 2\alpha.$$

$$C_{12} = 11 - 2\alpha, \quad C_{32} = 31 - 9\alpha, \quad C_{22} = 11\alpha - 5.$$

Question 6. Compute the determinant of the given matrix. If the matrix is invertible, find its inverse using the formula for 2×2 matrices.

1. $\begin{pmatrix} 3 & 5 \\ -2 & 4 \end{pmatrix}$

2. $\begin{pmatrix} 4 & 1 \\ 8 & 2 \end{pmatrix}$

3. $\begin{pmatrix} -5 & 7 \\ -7 & -2 \end{pmatrix}$

4. $\begin{pmatrix} \sqrt{2} & \sqrt{6} \\ 4 & \sqrt{3} \end{pmatrix}$

Solution.

1. The determinant of the matrix is

$$3 \times 4 - 5 \times (-2) = 12 + 10 = 22.$$

The inverse of the matrix is

$$\frac{1}{22} \begin{pmatrix} 4 & -5 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} \frac{2}{11} & -\frac{5}{22} \\ \frac{1}{11} & \frac{3}{22} \end{pmatrix}.$$

2. The matrix is not invertible

$$\begin{vmatrix} 4 & 1 \\ 8 & 2 \end{vmatrix} = 2 \times 4 - 8 \times 1 = 0$$

- 3.

$$\begin{vmatrix} -5 & 7 \\ -7 & -2 \end{vmatrix} = 59, \quad \begin{pmatrix} -5 & 7 \\ -7 & -2 \end{pmatrix}^{-1} = \begin{pmatrix} -\frac{2}{59} & -\frac{7}{59} \\ \frac{7}{59} & -\frac{5}{59} \end{pmatrix}$$

- 4.

$$\begin{vmatrix} \sqrt{2} & \sqrt{6} \\ 4 & \sqrt{3} \end{vmatrix} = -3\sqrt{6}, \quad \begin{pmatrix} \sqrt{2} & \sqrt{6} \\ 4 & \sqrt{3} \end{pmatrix}^{-1} = \begin{pmatrix} -\frac{\sqrt{2}}{6} & \frac{1}{3} \\ \frac{2\sqrt{6}}{9} & -\frac{\sqrt{3}}{9} \end{pmatrix}$$

Question 7. Find all values of λ for which $\det(A) = 0$.

$$1. A = \begin{pmatrix} \lambda - 2 & 1 \\ -5 & \lambda + 4 \end{pmatrix}$$

$$2. A = \begin{pmatrix} \lambda - 1 & 0 \\ 2 & \lambda + 1 \end{pmatrix}$$

$$3. A = \begin{pmatrix} \lambda - 4 & 0 & 0 \\ 0 & \lambda & 2 \\ 0 & 3 & \lambda - 1 \end{pmatrix}$$

$$4. A = \begin{pmatrix} \lambda - 4 & 4 & 0 \\ -1 & \lambda & 0 \\ 0 & 0 & \lambda - 5 \end{pmatrix}$$

$$5. A = \begin{pmatrix} \lambda & 2 \\ 5 & \lambda + 3 \end{pmatrix}$$

$$6. A = \begin{pmatrix} 15 & \lambda - 4 \\ \lambda + 7 & -2 \end{pmatrix}$$

$$7. A = \begin{pmatrix} \lambda - 3 & 5 & -19 \\ 0 & \lambda - 1 & 6 \\ 0 & 0 & \lambda - 2 \end{pmatrix}$$

Solution.

1.

$$\det(A) = (\lambda - 2)(\lambda + 4) + 5 = \lambda^2 + 2\lambda - 3, \quad \det(A) = 0 \iff \lambda = 1, -3$$

2.

$$\det(A) = (\lambda + 1)(\lambda - 1), \quad \det(A) = 0 \iff \lambda = 1, -1$$

3.

$$\det(A) = (\lambda - 4) \begin{vmatrix} \lambda & 2 \\ 3 & \lambda - 1 \end{vmatrix} = (\lambda - 4)(\lambda^2 - \lambda - 6), \quad \det(A) = 0 \iff \lambda = -2, 3, 4$$

4.

$$\det(A) = (\lambda - 5) \begin{vmatrix} \lambda - 4 & 4 \\ -1 & \lambda \end{vmatrix} = (\lambda - 5)(\lambda^2 - 4\lambda + 4), \quad \det(A) = 0 \iff \lambda = 2, 5$$

5.

$$\det(A) = \lambda(\lambda + 3) - 10 = \lambda^2 + 3\lambda - 10, \quad \det(A) = 0 \iff \lambda = -5, 2$$

6.

$$\det(A) = -\lambda^2 - 3\lambda - 2, \quad \det(A) = 0 \iff \lambda = -2, -1$$

7.

$$\det(A) = (\lambda - 3)(\lambda - 1)(\lambda - 2), \quad \det(A) = 0 \iff \lambda = 1, 2, 3$$

Question 8. Let

$$A = \begin{pmatrix} 3 & -2 & 4 \\ 5 & 1 & -2 \\ -1 & 3 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{pmatrix}.$$

Compute the determinant of matrices A, B by a cofactor expansion along

1. the first row

2. the second row

3. the third row

4. the first column

5. the second column

6. the third column

Solution.

1. Along the first row

$$\det(A) = 3 \begin{vmatrix} 1 & -2 \\ 3 & 6 \end{vmatrix} + 2 \begin{vmatrix} 5 & -2 \\ -1 & 6 \end{vmatrix} + 4 \begin{vmatrix} 5 & 1 \\ -1 & 3 \end{vmatrix} = 3 \times 12 + 2 \times 28 + 4 \times 16 = 156.$$

$$\det(B) = - \begin{vmatrix} 0 & -5 \\ 7 & 2 \end{vmatrix} - \begin{vmatrix} 3 & -5 \\ 1 & 2 \end{vmatrix} + 2 \begin{vmatrix} 3 & 0 \\ 1 & 7 \end{vmatrix} = -35 - 11 + 42 = -4.$$

2. Along the second row

$$\det(A) = 5 \times 24 + 1 \times 22 + (-2) \times (-7) = 156.$$

$$\det(B) = 3 \times 12 + 0 + (-5) \times 8 = -4.$$

3. Along the third row

$$\det(A) = (-1) \times 0 + 3 \times 26 + 6 \times 13 = 156.$$

$$\det(B) = 1 \times (-5) + 7 \times 1 + 2 \times (-3) = -4.$$

4. Along the first column

$$\det(A) = 3 \times 12 + 5 \times 24 + (-1) \times 0 = 156.$$

$$\det(B) = (-1) \times 35 + 3 \times 12 + 1 \times (-5) = -4.$$

5. Along the second column

$$\det(A) = (-2) \times (-28) + 1 \times 22 + 3 \times 26 = 156.$$

$$\det(B) = 1 \times (-11) + 0 + 7 \times 1 = -4.$$

6. Along the third column

$$\det(A) = 4 \times 16 + (-2) \times (-7) + 6 \times 13 = 156.$$

$$\det(B) = 2 \times 21 + (-5) \times 8 + 2 \times (-3) = -4.$$

Question 9. Compute the determinant of the following matrices by a cofactor expansion along a suitable row or column

$$1. \begin{pmatrix} -3 & 0 & 7 \\ 2 & 5 & 1 \\ -1 & 0 & 5 \end{pmatrix}$$

$$2. \begin{pmatrix} 3 & 3 & 1 \\ 1 & 0 & -4 \\ 1 & -3 & 5 \end{pmatrix}$$

$$3. \begin{pmatrix} k+1 & k-1 & 7 \\ 2 & k-3 & 4 \\ 5 & k+1 & k \end{pmatrix}$$

$$4. \begin{pmatrix} 5 & 2 & 1 & 0 \\ -1 & 3 & 5 & 2 \\ 4 & 1 & 0 & 2 \\ 0 & 2 & 3 & 0 \end{pmatrix}$$

$$5. \begin{pmatrix} 0 & 5 & 4 & 0 \\ 4 & 1 & -2 & 7 \\ -1 & 0 & 3 & 0 \\ 0 & 2 & 1 & 5 \end{pmatrix}$$

$$6. \begin{pmatrix} 2 & 1 & 9 & 7 \\ 0 & -1 & 3 & 8 \\ 0 & 0 & 5 & 2 \\ 0 & 0 & 0 & 6 \end{pmatrix}$$

$$7. \begin{pmatrix} 3 & 3 & 0 & 5 \\ 2 & -2 & 0 & -2 \\ 4 & 1 & -3 & 0 \\ 2 & 10 & 3 & 2 \end{pmatrix}$$

$$8. \begin{pmatrix} 0 & 4 & 1 & 3 & -2 \\ 2 & 2 & 3 & -1 & 0 \\ 3 & 1 & 2 & -5 & 1 \\ 1 & 0 & -4 & 0 & 0 \\ 0 & 3 & 0 & 0 & 2 \end{pmatrix}$$

$$9. \begin{pmatrix} 0 & 2 & 3 & 4 & -1 \\ 0 & 1 & 0 & 0 & -3 \\ 1 & 4 & 2 & 0 & -3 \\ 1 & 0 & 2 & 1 & 2 \\ 0 & 0 & 1 & -2 & 0 \end{pmatrix}$$

Solution.

1. Along the second column

$$\begin{vmatrix} -3 & 0 & 7 \\ 2 & 5 & 1 \\ -1 & 0 & 5 \end{vmatrix} = 5 \begin{vmatrix} -3 & 7 \\ -1 & 5 \end{vmatrix} = 5 \times (-8) = -40$$

2. Along the second row

$$1 \times (-18) + 0 + (-4) \times 12 = -66$$

3. Along the third column

$$7(17 - 3k) + 4(-k^2 + 3k - 6) + k(k^2 - 4k - 1) = k^3 - 8k^2 - 10k + 95$$

4. Along the fourth column

$$0 + 2 \times (-1) + 2 \times 1 + 0 = 0$$

5. Along the fourth column

$$0 + 7 \times (-3) + 0 + 5 \times (-46) = -251$$

6. Along the fourth row

$$6 \times (-10) = -60$$

7. Along the third column

$$0 + 0 + (-3) \times 144 + 3 \times (-32) = -528$$

8. Along the fourth row

$$1 \times 128 + 0 + (-4) \times (-56) + 0 = 352$$

9. Along the second row

$$1 \times (-51) + (-3) \times (-42) = 75$$

Question 10. Evaluate the determinant of the given matrix by inspection.

1. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

2. $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

3. $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 4 & 3 & 0 \\ 1 & 2 & 3 & 8 \end{pmatrix}$

4. $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{pmatrix}$

5. $\begin{pmatrix} 1 & 2 & 7 & -3 \\ 0 & 1 & -4 & 1 \\ 0 & 0 & 2 & 7 \\ 0 & 0 & 0 & 3 \end{pmatrix}$

6. $\begin{pmatrix} -3 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 40 & 10 & -1 & 0 \\ 100 & 200 & -23 & 3 \end{pmatrix}$

Solution.

1. -1 , 2. 8 , 3. 0 , 4. 24 , 5. 6 , 6. 18

Question 11. For each matrix, show that the value of the determinant is independent of θ .

1. $\begin{pmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{pmatrix}$

2. $\begin{pmatrix} \sin \theta & \cos \theta & 0 \\ -\cos \theta & \sin \theta & 0 \\ \sin \theta - \cos \theta & \sin \theta + \cos \theta & 1 \end{pmatrix}$

Solution.

1.

$$\begin{vmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{vmatrix} = \sin^2 \theta + \cos^2 \theta = 1.$$

2.

$$\begin{vmatrix} \sin \theta & \cos \theta & 0 \\ -\cos \theta & \sin \theta & 0 \\ \sin \theta - \cos \theta & \sin \theta + \cos \theta & 1 \end{vmatrix} = \begin{vmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{vmatrix} = 1$$

Question 12. By inspection, what is the relationship between the following determinants?

$$d_1 = \begin{vmatrix} a & b & c \\ d & 1 & f \\ g & 0 & 1 \end{vmatrix}, \quad d_2 = \begin{vmatrix} a + \lambda & b & c \\ d & 1 & f \\ g & 0 & 1 \end{vmatrix}$$

Solution. Let C_{12}, C_{13} denote the cofactors for the first matrix corresponding to entry $(1, 2)$ and $(1, 3)$, i.e. b and c . Then

$$d_1 = a + bC_{12} + cC_{13}, \quad d_2 = (a + \lambda) + bC_{12} + cC_{13}.$$

Thus

$$d_2 = d_1 + \lambda.$$

Question 13. Show that

$$\det(A) = \frac{1}{2} \begin{vmatrix} \operatorname{tr}(A) & 1 \\ \operatorname{tr}(A^2) & \operatorname{tr}(A) \end{vmatrix}$$

for every matrix $A \in \mathcal{M}_{2 \times 2}$.

Solution. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Then $\det(A) = ad - bc$, and

$$A^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ac + dc & bc + d^2 \end{pmatrix} \implies \operatorname{tr}(A^2) = a^2 + bc + bc + d^2 = a^2 + d^2 + 2bc$$

We have

$$\begin{aligned} \frac{1}{2} \begin{vmatrix} \operatorname{tr}(A) & 1 \\ \operatorname{tr}(A^2) & \operatorname{tr}(A) \end{vmatrix} &= \frac{1}{2} (\operatorname{tr}(A)^2 - \operatorname{tr}(A^2)) = \frac{1}{2} ((a + d)^2 - (a^2 + d^2 + 2bc)) \\ &= \frac{1}{2} (2ad - 2bc) = ad - bc = \det(A). \end{aligned}$$

Question 14. What can you say about $\det(A)$, where $A \in \mathcal{M}_{n \times n}$ has entries all equal to 1?

Solution.

For $n = 1$, $\det(A) = 1$.

For $n \geq 2$, the matrix A has all rows identical, hence $\det(A) = 0$.

Question 15. What is the maximum number of zeros that a 3×3 matrix can have without having a zero determinant? Why?

Solution. We note that I_3 has 6 zero entries and nonzero determinant.

If a 3×3 matrix A has more than 6 zero entries, it must have one row or one column entirely zero, making its determinant 0.

Hence the answer is 6.

Question 16. Explain why the determinant of a square matrix with integer entries must be an integer.

Solution. If we consider the cofactor expansion for computing the determinant, the determinant is always a sum of products of matrix entries and their minors (which are themselves determinants of smaller matrices).

The determinant calculation involves addition, subtraction, and multiplication, all of which preserve integer values when applied to integers.

Question 17. Prove that the three points (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) are collinear if and only if

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

Solution. We prove both directions.

Assume first that the three points (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) are collinear. Then they lie on the same line, so the slopes between any two pairs of distinct points are equal. Hence

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_1}{x_3 - x_1}.$$

Cross-multiplying gives

$$(x_2 - x_1)(y_3 - y_1) = (x_3 - x_1)(y_2 - y_1).$$

Expanding both sides,

$$x_2y_3 - x_2y_1 - x_1y_3 + x_1y_1 = x_3y_2 - x_3y_1 - x_1y_2 + x_1y_1,$$

so

$$x_2y_3 - x_3y_2 - x_1y_3 + x_3y_1 + x_1y_2 - x_2y_1 = 0.$$

Now compute the determinant by cofactor expansion along the third column:

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} - \begin{vmatrix} x_1 & y_1 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}.$$

Thus

$$\begin{aligned} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} &= (x_2y_3 - x_3y_2) - (x_1y_3 - x_3y_1) + (x_1y_2 - x_2y_1) \\ &= x_2y_3 - x_3y_2 - x_1y_3 + x_3y_1 + x_1y_2 - x_2y_1 \\ &= 0. \end{aligned}$$

Conversely, assume that

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

Expanding along the third column gives

$$x_2y_3 - x_3y_2 - x_1y_3 + x_3y_1 + x_1y_2 - x_2y_1 = 0.$$

Rearranging,

$$(x_2 - x_1)(y_3 - y_1) = (x_3 - x_1)(y_2 - y_1).$$

Therefore

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_1}{x_3 - x_1},$$

whenever the denominators are nonzero. Hence the three points have the same slope and are collinear.

If $x_2 = x_1$, then

$$0 = (x_3 - x_1)(y_2 - y_1).$$

If $y_2 \neq y_1$, then $x_3 = x_1$, so all three points lie on the same vertical line. If $y_2 = y_1$, then the first two points coincide. In that case, any line through this common point and the third point contains all three points, so the three points are collinear. The other cases in which two points have the same x -coordinate are analogous.

Question 18. Prove that the equation of the line through the distinct points (a_1, b_1) and (a_2, b_2) can be written as

$$\begin{vmatrix} x & y & 1 \\ a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \end{vmatrix} = 0$$

Solution. The line through (a_1, b_1) and (a_2, b_2) is

$$(a_1 - a_2)y + (b_2 - b_1)x = a_1b_2 - a_2b_1.$$

Cofactor expansion along the first row for computing the determinant gives

$$\begin{vmatrix} x & y & 1 \\ a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \end{vmatrix} = x \begin{vmatrix} b_1 & 1 \\ b_2 & 1 \end{vmatrix} - y \begin{vmatrix} a_1 & 1 \\ a_2 & 1 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = x(b_1 - b_2) - y(a_1 - a_2) + (a_1b_2 - a_2b_1) = 0$$

i.e.

$$(a_1 - a_2)y + (b_2 - b_1)x = a_1b_2 - a_2b_1.$$

Question 19. Prove

Theorem 1 *A triangular matrix is invertible if and only if its diagonal entries are all nonzero.*

Solution. Let $A = (a_{ij}) \in \mathcal{M}_{n \times n}$ be a triangular matrix, then its determinant is given by

$$\det(A) = a_{11}a_{22} \cdots a_{nn}$$

We have proved during the lecture that a matrix is invertible if and only if its determinant is nonzero. Thus A is invertible iff the diagonal entries are all nonzero.

Question 20. Prove Theorem 3 using Theorems 1 and 2.

Theorem 2 *If A is an invertible matrix, then*

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Theorem 3 *The inverse of an invertible lower triangular matrix is lower triangular, and the inverse of an invertible upper triangular matrix is upper triangular.*

Solution. Assume $A \in \mathcal{M}_{n \times n}$ is upper triangular and invertible. By Theorem 2,

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Thus, it suffices to show that $\operatorname{adj}(A)$ is upper triangular. Equivalently, it suffices to show that the matrix of cofactors of A is lower triangular.

To prove this, we show that $C_{ij} = 0$ whenever $i < j$. Since

$$C_{ij} = (-1)^{i+j} M_{ij},$$

it is enough to prove that $M_{ij} = 0$ whenever $i < j$.

Let B_{ij} be the matrix obtained from A by deleting the i th row and the j th column, where $i < j$. Then

$$M_{ij} = \det(B_{ij}).$$

Because A is upper triangular and $i < j$, the matrix B_{ij} is also upper triangular. Moreover, the i th row of B_{ij} comes from the $(i+1)$ th row of A , which begins with at least i zeros. Since the deleted column satisfies $j > i$, none of these zeros is removed, so the (i, i) -entry of B_{ij} is 0.

Therefore B_{ij} is an upper triangular matrix with a zero entry on its main diagonal. By Theorem 1, B_{ij} is not invertible, so

$$M_{ij} = \det(B_{ij}) = 0.$$

Hence $C_{ij} = 0$ for all $i < j$, which shows that the cofactor matrix is lower triangular. Therefore $\operatorname{adj}(A)$, being its transpose, is upper triangular. It follows from Theorem 2 that A^{-1} is upper triangular.

The proof for the lower triangular case is similar.

Question 21. With the following examples, verify that $\det(A) = \det(A^\top)$

$$1. A = \begin{pmatrix} -2 & 3 \\ 1 & 4 \end{pmatrix}$$

$$2. A = \begin{pmatrix} -6 & 1 \\ 2 & -2 \end{pmatrix}$$

$$3. A = \begin{pmatrix} 2 & -1 & 3 \\ 1 & 2 & 4 \\ 5 & -3 & 6 \end{pmatrix}$$

$$4. A = \begin{pmatrix} 4 & 2 & -1 \\ 0 & 2 & -3 \\ -1 & 1 & 5 \end{pmatrix}$$

Solution.

$$1. \det(A) = \det(A^\top) = -11$$

$$2. \det(A) = \det(A^\top) = 10$$

$$3. \det(A) = \det(A^\top) = -5$$

$$4. \det(A) = \det(A^\top) = 56$$

Question 22. Each of the following matrices is obtained from the identity matrix by performing a single elementary row operation. Identify the specific operation performed and find the determinant of each matrix using the properties of determinants in relation to row operations.

$$1. \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$2. \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$3. \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{pmatrix}$$

$$4. \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$5. \begin{pmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$6. \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$7. \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$8. \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$9. \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -4 \end{pmatrix}$$

$$10. \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Solution.

1. This elementary matrix is obtained by multiplying the third row of I_4 by -5 , which, according to determinant properties, results in a determinant of -5
2. The elementary matrix is obtained by interchanging the second and third rows of I_4 , which results in a determinant of -1 .
3. The elementary matrix is obtained by adding -5 times the first row to the third row of I_3 , yielding a determinant of 1 .
4. The elementary matrix is obtained by multiplying the second row of I_4 by $-\frac{1}{3}$, which gives a determinant of $-\frac{1}{3}$.
5. $R_1 \rightarrow R_1 - 3R_2$, determinant $= 1$.
6. $R_2 \rightarrow R_2 + 2R_1$, determinant $= 1$.
7. $R_1 \rightarrow 1R_1$, determinant $= 1$.
8. $R_1 \rightarrow \frac{1}{2}R_1$, determinant $= \frac{1}{2}$.
9. $R_3 \rightarrow -4R_3$, determinant $= -4$.
10. $R_1 \leftrightarrow R_2$, determinant $= -1$.

Question 23. Find the determinant of the following matrices using some combination of row operations, column operations, and cofactor expansion.

$$1. \begin{pmatrix} 10 & 4 & 21 \\ 0 & -4 & 3 \\ -5 & -1 & -12 \end{pmatrix}$$

$$2. \begin{pmatrix} 18 & -9 & -14 \\ 6 & -3 & -5 \\ -3 & 1 & 2 \end{pmatrix}$$

$$3. \begin{pmatrix} 3 & -6 & 9 \\ -2 & 7 & -2 \\ 0 & 1 & 5 \end{pmatrix}$$

$$4. \begin{pmatrix} 3 & 6 & -9 \\ 0 & 0 & -2 \\ -2 & 1 & 5 \end{pmatrix}$$

$$5. \begin{pmatrix} 1 & -3 & 0 \\ -2 & 4 & 1 \\ 5 & -2 & 2 \end{pmatrix}$$

$$6. \begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ -2 & -7 & 0 & -4 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

$$7. \begin{pmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 8 & 6 & 1 \end{pmatrix}$$

$$8. \begin{pmatrix} -8 & 4 & -3 & 2 \\ 2 & 1 & -1 & -1 \\ -3 & -5 & 4 & 0 \\ 2 & -4 & 3 & -1 \end{pmatrix}$$

$$9. \begin{pmatrix} 1 & -1 & 5 & 1 \\ -2 & 1 & -7 & 1 \\ -3 & 2 & -12 & -2 \\ 2 & -1 & 9 & 1 \end{pmatrix}$$

$$10. \begin{pmatrix} 5 & 3 & -8 & 4 \\ \frac{15}{2} & \frac{1}{2} & -1 & -7 \\ -\frac{5}{2} & \frac{3}{2} & -4 & 1 \\ 10 & -3 & 8 & -8 \end{pmatrix}$$

$$11. \begin{pmatrix} 1 & 2 & -1 & 3 & 0 \\ 2 & 4 & -3 & 1 & -4 \\ 2 & 6 & 4 & 8 & -4 \\ -3 & -8 & -1 & 1 & 0 \\ 1 & 3 & 3 & 10 & 1 \end{pmatrix}$$

$$12. \begin{pmatrix} -4 & 1 & 1 & 1 & 1 \\ 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & -4 \end{pmatrix}$$

Solution.

1.

$$\begin{vmatrix} 10 & 4 & 21 \\ 0 & -4 & 3 \\ -5 & -1 & -12 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 + \frac{1}{2}R_1} \begin{vmatrix} 10 & 4 & 21 \\ 0 & -4 & 3 \\ 0 & 1 & -\frac{3}{2} \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 + 4R_3} \begin{vmatrix} 10 & 4 & 21 \\ 0 & 0 & -3 \\ 0 & 1 & -\frac{3}{2} \end{vmatrix} \xrightarrow{R_2 \leftrightarrow R_3} - \begin{vmatrix} 10 & 4 & 21 \\ 0 & 1 & -\frac{3}{2} \\ 0 & 0 & -3 \end{vmatrix} = 30$$

2.

$$\begin{vmatrix} 18 & -9 & -14 \\ 6 & -3 & -5 \\ -3 & 1 & 2 \end{vmatrix} \xrightarrow{R_1 \rightarrow R_1 - 3R_2} \begin{vmatrix} 0 & 0 & 1 \\ 6 & -3 & -5 \\ -3 & 1 & 2 \end{vmatrix} = \begin{vmatrix} 6 & -3 \\ -3 & 1 \end{vmatrix} = 6 - 9 = -3$$

3.

$$\begin{vmatrix} 3 & -6 & 9 \\ -2 & 7 & -2 \\ 0 & 1 & 5 \end{vmatrix} \xrightarrow{R_1 \rightarrow R_1 + 6R_3, R_2 \rightarrow R_2 - 7R_3} \begin{vmatrix} 3 & 0 & 39 \\ -2 & 0 & -37 \\ 0 & 1 & 5 \end{vmatrix} = - \begin{vmatrix} 3 & 39 \\ -2 & -37 \end{vmatrix} = 33$$

4.

$$\begin{vmatrix} 3 & 6 & -9 \\ 0 & 0 & -2 \\ -2 & 1 & 5 \end{vmatrix} = 2 \begin{vmatrix} 3 & 6 \\ -2 & 1 \end{vmatrix} = 2 \times (3 + 12) = 30$$

5.

$$\begin{vmatrix} 1 & -3 & 0 \\ -2 & 4 & 1 \\ 5 & -2 & 2 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 + 2R_1, R_3 \rightarrow R_3 - 5R_1} \begin{vmatrix} 1 & -3 & 0 \\ 0 & -2 & 1 \\ 0 & 13 & 2 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 + \frac{13}{2}R_2} \begin{vmatrix} 1 & -3 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & \frac{17}{2} \end{vmatrix} = -17$$

6.

$$\begin{vmatrix} 1 & 3 & 1 & 5 & 3 \\ -2 & -7 & 0 & -4 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 + 2R_1} \begin{vmatrix} 1 & 3 & 1 & 5 & 3 \\ 0 & -1 & 2 & 6 & 8 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{vmatrix} \xrightarrow{R_4 \rightarrow R_4 - 2R_3} \begin{vmatrix} 1 & 3 & 1 & 5 & 3 \\ 0 & -1 & 2 & 6 & 8 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{vmatrix}$$

$$\xrightarrow{R_5 \rightarrow R_5 - R_4} \begin{vmatrix} 1 & 3 & 1 & 5 & 3 \\ 0 & -1 & 2 & 6 & 8 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{vmatrix} = -2$$

7.

$$\begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 8 & 6 & 1 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - 5R_1, R_3 \rightarrow R_3 + R_1, R_4 \rightarrow R_4 - 2R_1} \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 12 & 0 & -1 \end{vmatrix} \xrightarrow{R_4 \rightarrow R_4 - 12R_2} \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 108 & 23 \end{vmatrix}$$

$$\xrightarrow{R_4 \rightarrow R_4 + 36R_3} \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 0 & -13 \end{vmatrix} = 39$$

8.

$$\begin{vmatrix} -8 & 4 & -3 & 2 \\ 2 & 1 & -1 & -1 \\ -3 & -5 & 4 & 0 \\ 2 & -4 & 3 & -1 \end{vmatrix} \xrightarrow{R_1 \rightarrow R_1 + 4R_2, R_4 \rightarrow R_4 - R_2, R_3 \rightarrow R_3 + R_2} \begin{vmatrix} 0 & 8 & -7 & -2 \\ 2 & 1 & -1 & -1 \\ -1 & -4 & 3 & -1 \\ 0 & -5 & 4 & 0 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 + 2R_3} \begin{vmatrix} 0 & 8 & -7 & -2 \\ 0 & 1 & -2 & -5 \\ -1 & -4 & 3 & -1 \\ 0 & -5 & 4 & 0 \end{vmatrix}$$

$$\xrightarrow{R_1 \rightarrow R_1 + R_2} \begin{vmatrix} 0 & 8 & -7 & -2 \\ 0 & 1 & -2 & -5 \\ -1 & -4 & 3 & -1 \\ 0 & -5 & 4 & 0 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 + 7R_1, R_4 \rightarrow R_4 + 5R_1} \begin{vmatrix} 0 & 1 & -2 & -5 \\ 0 & 0 & -9 & -38 \\ -1 & -4 & 3 & -1 \\ 0 & 0 & -6 & -25 \end{vmatrix}$$

$$\xrightarrow{R_1 \leftrightarrow R_3} - \begin{vmatrix} -1 & -4 & 3 & -1 \\ 0 & 0 & -9 & -38 \\ 0 & 1 & -2 & -5 \\ 0 & 0 & -6 & -25 \end{vmatrix} \xrightarrow{R_2 \leftrightarrow R_3} - \begin{vmatrix} -1 & -4 & 3 & -1 \\ 0 & 1 & -2 & -5 \\ 0 & 0 & -9 & -38 \\ 0 & 0 & -6 & -25 \end{vmatrix} = - \begin{vmatrix} 1 & -2 & -5 \\ 0 & -9 & -38 \\ 0 & -6 & -25 \end{vmatrix} = - \begin{vmatrix} -9 & -38 \\ -6 & -25 \end{vmatrix} = 3$$

9.

$$\begin{vmatrix} 1 & -1 & 5 & 1 \\ -2 & 1 & -7 & 1 \\ -3 & 2 & -12 & -2 \\ 2 & -1 & 9 & 1 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 + 2R_1, R_3 \rightarrow R_3 + 3R_1, R_4 \rightarrow R_4 - 2R_1} \begin{vmatrix} 1 & -1 & 5 & 1 \\ 0 & -1 & 3 & 3 \\ 0 & -1 & 3 & 1 \\ 0 & 1 & -1 & -1 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 + R_2} \begin{vmatrix} 1 & -1 & 5 & 1 \\ 0 & -1 & 3 & 3 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 2 \end{vmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} - \begin{vmatrix} 1 & -1 & 5 & 1 \\ 0 & -1 & 3 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & -2 \end{vmatrix} = -4$$

10.

$$\begin{vmatrix} 5 & 3 & -8 & 4 \\ \frac{15}{2} & \frac{1}{2} & -1 & -7 \\ -\frac{5}{2} & \frac{3}{2} & -4 & 1 \\ 10 & -3 & 8 & -8 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 + 3R_3, R_4 \rightarrow R_4 - 2R_1} \begin{vmatrix} 5 & 3 & -8 & 4 \\ 0 & 5 & -13 & -4 \\ -\frac{5}{2} & \frac{3}{2} & -4 & 1 \\ 0 & -9 & 24 & -16 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 + \frac{1}{2}R_1} \begin{vmatrix} 5 & 3 & -8 & 4 \\ 0 & 5 & -13 & -4 \\ 0 & 3 & -8 & 3 \\ 0 & -9 & 24 & -16 \end{vmatrix}$$

$$\xrightarrow{R_4 \rightarrow R_4 + 3R_3} \begin{vmatrix} 5 & 3 & -8 & 4 \\ 0 & 5 & -13 & -4 \\ 0 & 3 & -8 & 3 \\ 0 & 0 & 0 & -7 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 - \frac{3}{5}R_2} \begin{vmatrix} 5 & 3 & -8 & 4 \\ 0 & 5 & -13 & -4 \\ 0 & 0 & -\frac{1}{5} & \frac{27}{5} \\ 0 & 0 & 0 & -7 \end{vmatrix} = 35$$

11.

$$\begin{vmatrix} 1 & 2 & -1 & 3 & 0 \\ 2 & 4 & -3 & 1 & -4 \\ 2 & 6 & 4 & 8 & -4 \\ -3 & -8 & -1 & 1 & 0 \\ 1 & 3 & 3 & 10 & 1 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 2R_1, R_4 \rightarrow R_4 + 3R_1, R_5 \rightarrow R_5 - R_1} \begin{vmatrix} 1 & 2 & -1 & 3 & 0 \\ 0 & 0 & -1 & -5 & -4 \\ 0 & 2 & 6 & 2 & -4 \\ 0 & -2 & -4 & 10 & 0 \\ 0 & 1 & 4 & 7 & 1 \end{vmatrix}$$

$$\xrightarrow{R_3 \rightarrow R_3 + R_3, R_4 \rightarrow R_4 + 2R_5} \begin{vmatrix} 1 & 2 & -1 & 3 & 0 \\ 0 & 0 & -1 & -5 & -4 \\ 0 & 0 & 2 & 12 & -4 \\ 0 & 0 & 4 & 24 & 2 \\ 0 & 1 & 4 & 7 & 1 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 + 2R_2, R_4 \rightarrow R_4 + 4R_2} \begin{vmatrix} 1 & 2 & -1 & 3 & 0 \\ 0 & 0 & -1 & -5 & -4 \\ 0 & 0 & 0 & 2 & -12 \\ 0 & 0 & 0 & 4 & -14 \\ 0 & 1 & 4 & 7 & 1 \end{vmatrix}$$

$$\xrightarrow{R_5 \leftrightarrow R_2, R_4 \rightarrow R_4 - 2R_3} \begin{vmatrix} 1 & 2 & -1 & 3 & 0 \\ 0 & 1 & 4 & 7 & 1 \\ 0 & 0 & 0 & 2 & -12 \\ 0 & 0 & 0 & 0 & 10 \\ 0 & 0 & -1 & -5 & -4 \end{vmatrix} \xrightarrow{R_5 \leftrightarrow R_3, R_4 \leftrightarrow R_5} \begin{vmatrix} 1 & 2 & -1 & 3 & 0 \\ 0 & 1 & 4 & 7 & 1 \\ 0 & 0 & -1 & -5 & -4 \\ 0 & 0 & 0 & 2 & -12 \\ 0 & 0 & 0 & 0 & 10 \end{vmatrix} = 20$$

12.

$$\begin{vmatrix} -4 & 1 & 1 & 1 & 1 \\ 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & -4 \end{vmatrix} \xrightarrow{R_5 \rightarrow R_5 - R_4, R_4 \rightarrow R_4 - R_3, R_3 \rightarrow R_3 - R_2, R_2 \rightarrow R_2 - R_1} \begin{vmatrix} -4 & 1 & 1 & 1 & 1 \\ 5 & -5 & 0 & 0 & 0 \\ 0 & 5 & -5 & 0 & 0 \\ 0 & 0 & 5 & -5 & 0 \\ 0 & 0 & 0 & 5 & -5 \end{vmatrix}$$

$$= 5^4 \begin{vmatrix} -4 & 1 & 1 & 1 & 1 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{vmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_5} 5^4 \begin{vmatrix} -4 & 1 & 1 & 2 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{vmatrix} \xrightarrow{R_1 \rightarrow R_1 + 2R_4}$$

$$5^4 \begin{vmatrix} -4 & 1 & 3 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{vmatrix} \xrightarrow{R_1 \rightarrow R_1 + 3R_3} 5^4 \begin{vmatrix} -4 & 4 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{vmatrix} \xrightarrow{R_1 \rightarrow R_1 + 4R_2}$$

$$5^4 \begin{vmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{vmatrix} = 0$$

Question 24. Prove that if a square matrix has two proportional rows or columns, then its determinant is zero.

Solution. Let A be an $n \times n$ square matrix with two proportional rows, say $R_i = \alpha R_j$ for some scalar α . Consider performing the row operation:

$$A \xrightarrow{R_i \rightarrow R_i - \alpha R_j} B.$$

Since this operation replaces R_i with the zero row, the resulting matrix B has a row of zeros. It is well known that a matrix with a zero row has a determinant of zero, so we conclude that

$$\det(A) = \det(B) = 0.$$

A similar argument holds if A has two proportional columns.

Question 25. Compute the following determinants, given that

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$$

$$1. \begin{vmatrix} d & e & f \\ g & h & i \\ a & b & c \end{vmatrix}$$

$$2. \begin{vmatrix} g & h & i \\ d & e & f \\ a & b & c \end{vmatrix}$$

$$3. \begin{vmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ 4g & 4h & 4i \end{vmatrix}$$

$$4. \begin{vmatrix} a+d & b+e & c+f \\ -d & -e & -f \\ g & h & i \end{vmatrix}$$

$$5. \begin{vmatrix} a+g & b+h & c+i \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$6. \begin{vmatrix} a & b & c \\ d & e & f \\ 2a & 2b & 2c \end{vmatrix}$$

$$7. \begin{vmatrix} a & b & c \\ 2d & 2e & 2f \\ g+3a & h+3b & i+3c \end{vmatrix}$$

$$8. \begin{vmatrix} -3a & -3b & -3c \\ d & e & f \\ g-4d & h-4e & i-4f \end{vmatrix}$$

Solution.

1.

$$\begin{vmatrix} d & e & f \\ g & h & i \\ a & b & c \end{vmatrix} \xrightarrow{R_3 \leftrightarrow R_1} - \begin{vmatrix} a & b & c \\ g & h & i \\ d & e & f \end{vmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$$

2.

$$\begin{vmatrix} g & h & i \\ d & e & f \\ a & b & c \end{vmatrix} \stackrel{R_1 \leftrightarrow R_3}{=} - \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 6$$

3.

$$\begin{vmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ 4g & 4h & 4i \end{vmatrix} \stackrel{R_1 \rightarrow \frac{1}{3}R_1}{=} 3 \begin{vmatrix} a & b & c \\ -d & -e & -f \\ 4g & 4h & 4i \end{vmatrix} \stackrel{R_2 \rightarrow -R_2}{=} -3 \begin{vmatrix} a & b & c \\ d & e & f \\ 4g & 4h & 4i \end{vmatrix} \stackrel{R_3 \rightarrow \frac{1}{4}R_3}{=} -12 \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 72$$

4.

$$\begin{vmatrix} a+d & b+e & c+f \\ -d & -e & -f \\ g & h & i \end{vmatrix} = \begin{vmatrix} a & b & c \\ -d & -e & -f \\ g & h & i \end{vmatrix} + \begin{vmatrix} d & e & f \\ -d & -e & -f \\ g & h & i \end{vmatrix} = - \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + 0 = 6$$

5.

$$\begin{vmatrix} a+g & b+h & c+i \\ d & e & f \\ g & h & i \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + \begin{vmatrix} g & h & i \\ d & e & f \\ g & h & i \end{vmatrix} = -6 + 0 = -6$$

6.

$$\begin{vmatrix} a & b & c \\ d & e & f \\ 2a & 2b & 2c \end{vmatrix} = 0$$

7.

$$\begin{vmatrix} a & b & c \\ 2d & 2e & 2f \\ g+3a & h+3b & i+3c \end{vmatrix} = \begin{vmatrix} a & b & c \\ 2d & 2e & 2f \\ g & h & i \end{vmatrix} + \begin{vmatrix} a & b & c \\ 2d & 2e & 2f \\ 3a & 3b & 3c \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + 0 = -12$$

8.

$$\begin{vmatrix} -3a & -3b & -3c \\ d & e & f \\ g-4d & h-4e & i-4f \end{vmatrix} = \begin{vmatrix} -3a & -3b & -3c \\ d & e & f \\ g & h & i \end{vmatrix} + \begin{vmatrix} -3a & -3b & -3c \\ d & e & f \\ -4d & -4e & -4f \end{vmatrix} = -3 \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 18$$

Question 26. Use row reduction to show that

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$$

Solution.

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} \stackrel{R_2 \rightarrow R_2 - aR_1, R_3 \rightarrow R_3 - a^2R_1}{=} \begin{vmatrix} 1 & 1 & 1 \\ 0 & b-a & c-a \\ 0 & b^2-a^2 & c^2-a^2 \end{vmatrix} \stackrel{R_3 \rightarrow R_3 - (a+b)R_2}{=} \begin{vmatrix} 1 & 1 & 1 \\ 0 & b-a & c-a \\ 0 & 0 & (c^2-a^2) - (c-a)(a+b) \end{vmatrix} = (b-a)((c^2-a^2) - (c-a)(a+b)) = (b-a)(c-a)(c-b)$$

Question 27. Verify the following two formulas and make a conjecture about a general result of which these results are special cases

$$1. \begin{vmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = -a_{13}a_{22}a_{31} \quad 2. \begin{vmatrix} 0 & 0 & 0 & a_{14} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} = a_{14}a_{23}a_{32}a_{41}$$

Solution.

1.

$$\begin{vmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_3} - \begin{vmatrix} a_{31} & a_{32} & a_{33} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{13} \end{vmatrix} = -a_{13}a_{22}a_{31}$$

2.

$$\begin{vmatrix} 0 & 0 & 0 & a_{14} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_4} - \begin{vmatrix} a_{41} & a_{42} & a_{43} & a_{44} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{14} \end{vmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{vmatrix} a_{41} & a_{42} & a_{43} & a_{44} \\ 0 & a_{32} & a_{33} & a_{34} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & 0 & 0 & a_{14} \end{vmatrix} = a_{14}a_{23}a_{32}a_{41}$$

In general,

$$\begin{vmatrix} 0 & 0 & \cdots & 0 & a_{1n} \\ 0 & 0 & \cdots & a_{2(n-1)} & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{n(n-1)} & a_{nn} \end{vmatrix} = (-1)^\ell a_{n1}a_{(n-1)2} \cdots a_{2(n-1)}a_{1n},$$

where

$$\ell = \left\lfloor \frac{n}{2} \right\rfloor.$$

Question 28. Confirm the following identities

$$1. \begin{vmatrix} a_1 & b_1 & a_1 + b_1 + c_1 \\ a_2 & b_2 & a_2 + b_2 + c_2 \\ a_3 & b_3 & a_3 + b_3 + c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$2. \begin{vmatrix} a_1 + b_1 t & a_2 + b_2 t & a_3 + b_3 t \\ a_1 t + b_1 & a_2 t + b_2 & a_3 t + b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = (1 - t^2) \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$3. \begin{vmatrix} a_1 + b_1 & a_1 - b_1 & c_1 \\ a_2 + b_2 & a_2 - b_2 & c_2 \\ a_3 + b_3 & a_3 - b_3 & c_3 \end{vmatrix} = -2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$4. \begin{vmatrix} a_1 & b_1 + ta_1 & c_1 + rb_1 + sa_1 \\ a_2 & b_2 + ta_2 & c_2 + rb_2 + sa_2 \\ a_3 & b_3 + ta_3 & c_3 + rb_3 + sa_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Solution.

1.

$$\begin{vmatrix} a_1 & b_1 & a_1 + b_1 + c_1 \\ a_2 & b_2 & a_2 + b_2 + c_2 \\ a_3 & b_3 & a_3 + b_3 + c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & a_1 \\ a_2 & b_2 & a_2 \\ a_3 & b_3 & a_3 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & b_1 \\ a_2 & b_2 & b_2 \\ a_3 & b_3 & b_3 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

2.

$$\begin{aligned}
\begin{vmatrix} a_1 + b_1 t & a_2 + b_2 t & a_3 + b_3 t \\ a_1 t + b_1 & a_2 t + b_2 & a_3 t + b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} &= \begin{vmatrix} a_1 & a_2 & a_3 \\ a_1 t + b_1 & a_2 t + b_2 & a_3 t + b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} b_1 t & b_2 t & b_3 t \\ a_1 t + b_1 & a_2 t + b_2 & a_3 t + b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \\
&= \begin{vmatrix} a_1 & a_2 & a_3 \\ a_1 t & a_2 t & a_3 t \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} b_1 t & b_2 t & b_3 t \\ a_1 t & a_2 t & a_3 t \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} b_1 t & b_2 t & b_3 t \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \\
&= 0 + \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + t^2 \begin{vmatrix} b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + 0 = (1 - t^2) \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}
\end{aligned}$$

3.

$$\begin{aligned}
\begin{vmatrix} a_1 + b_1 & a_1 - b_1 & c_1 \\ a_2 + b_2 & a_2 - b_2 & c_2 \\ a_3 + b_3 & a_3 - b_3 & c_3 \end{vmatrix} &= \begin{vmatrix} a_1 & a_1 - b_1 & c_1 \\ a_2 & a_2 - b_2 & c_2 \\ a_3 & a_3 - b_3 & c_3 \end{vmatrix} + \begin{vmatrix} b_1 & a_1 - b_1 & c_1 \\ b_2 & a_2 - b_2 & c_2 \\ b_3 & a_3 - b_3 & c_3 \end{vmatrix} \\
&= \begin{vmatrix} a_1 & a_1 & c_1 \\ a_2 & a_2 & c_2 \\ a_3 & a_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & -b_1 & c_1 \\ a_2 & -b_2 & c_2 \\ a_3 & -b_3 & c_3 \end{vmatrix} + \begin{vmatrix} b_1 & a_1 & c_1 \\ b_2 & a_2 & c_2 \\ b_3 & a_3 & c_3 \end{vmatrix} + \begin{vmatrix} b_1 & -b_1 & c_1 \\ b_2 & -b_2 & c_2 \\ b_3 & -b_3 & c_3 \end{vmatrix} \\
&= 0 - \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} - \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + 0 = -2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}
\end{aligned}$$

4.

$$\begin{aligned}
\begin{vmatrix} a_1 & b_1 + ta_1 & c_1 + rb_1 + sa_1 \\ a_2 & b_2 + ta_2 & c_2 + rb_2 + sa_2 \\ a_3 & b_3 + ta_3 & c_3 + rb_3 + sa_3 \end{vmatrix} &= \begin{vmatrix} a_1 & b_1 & c_1 + rb_1 + sa_1 \\ a_2 & b_2 & c_2 + rb_2 + sa_2 \\ a_3 & b_3 & c_3 + rb_3 + sa_3 \end{vmatrix} + \begin{vmatrix} a_1 & ta_1 & c_1 + rb_1 + sa_1 \\ a_2 & ta_2 & c_2 + rb_2 + sa_2 \\ a_3 & ta_3 & c_3 + rb_3 + sa_3 \end{vmatrix} \\
&= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & rb_1 \\ a_2 & b_2 & rb_2 \\ a_3 & b_3 & rb_3 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & sa_1 \\ a_2 & b_2 & sa_2 \\ a_3 & b_3 & sa_3 \end{vmatrix} + 0 \\
&= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}
\end{aligned}$$

Question 29. For the following matrices, show that $\det(A) = 0$ without explicitly computing the determinant.

$$1. A = \begin{pmatrix} -2 & 8 & 1 & 4 \\ 3 & 2 & 5 & 1 \\ 1 & 10 & 6 & 5 \\ 4 & -6 & 4 & -3 \end{pmatrix}$$

$$2. A = \begin{pmatrix} -4 & 1 & 1 & 1 & 1 \\ 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & -4 \end{pmatrix}$$

Solution.

1.

$$\det(A) \xrightarrow{R_1 \rightarrow R_1 + R_2 - R_3} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 3 & 2 & 5 & 1 \\ 1 & 10 & 6 & 5 \\ 4 & -6 & 4 & -3 \end{pmatrix}$$

2.

$$\det(A)^{R_1 \rightarrow R_1 + R_2 + R_3 + R_4 + R_5} = \begin{vmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & -4 \end{vmatrix} = 0$$

Question 30. It can be proved that if a square matrix M can be partitioned into *block triangular form* as

$$M = \begin{pmatrix} A & 0 \\ C & B \end{pmatrix} \quad \text{or} \quad M = \begin{pmatrix} A & C \\ 0 & B \end{pmatrix}$$

where A and B are square matrices, then

$$\det(M) = \det(A) \det(B).$$

Use this result to compute the determinants of the following matrices:

$$1. \ M = \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 8 & 6 & -9 \\ 2 & 5 & 0 & 4 & 7 & 5 \\ -1 & 3 & 2 & 6 & 9 & -2 \\ \hline 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & -3 & 8 & -4 \end{array} \right) \quad 2. \ M = \left(\begin{array}{ccc|cc} 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 & 2 \\ 2 & 0 & 0 & 0 & 1 \end{array} \right)$$

Solution.

1.

$$\det(M) = \begin{vmatrix} 1 & 2 & 0 \\ 2 & 5 & 0 \\ -1 & 3 & 2 \end{vmatrix} \begin{vmatrix} 3 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 8 & -4 \end{vmatrix} = \left(\begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} | 2| \right) \left(\begin{vmatrix} 3 & 0 \\ 2 & 1 \end{vmatrix} | -4| \right) = 2 \times (-12) = -24$$

2.

$$\det(M) = \begin{vmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1 \times 1 = 1$$

Question 31. Given $A \in \mathcal{M}_{n \times n}$. Let B be the matrix that results when the rows of A are written in reverse order. Formulate a statement that describes the relationship between $\det(A)$ and $\det(B)$.

Solution. Reversing the rows of A can be seen as performing a sequence of row swaps. Specifically, we swap:

- The first row with the last row,
- The second row with the second-last row,
- And so on.

Thus, we can obtain B from A by repeatedly applying the row operation that swaps two rows. The number of such operations needed is

$$\left\lfloor \frac{n}{2} \right\rfloor.$$

We have

$$\det(B) = (-1)^{\left\lfloor \frac{n}{2} \right\rfloor} \det(A).$$

Question 32. Find the determinant of the following matrix.

$$\begin{pmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{pmatrix}$$

Solution.

$$\begin{aligned} & \begin{vmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{vmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2, \ R_2 \rightarrow \underline{\underline{R_2 - R_3}}, \ R_3 \rightarrow R_3 - R_4} \begin{vmatrix} a-b & b-a & 0 & 0 \\ 0 & a-b & b-a & 0 \\ 0 & 0 & a-b & b-a \\ b & b & b & a \end{vmatrix} = (a-b)^3 \begin{vmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ b & b & b & a \end{vmatrix} \\ & \xrightarrow{R_4 \rightarrow \underline{\underline{R_4 - bR_1}}} (a-b)^3 \begin{vmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 2b & b & a \end{vmatrix} = (a-b)^3 \begin{vmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 3b & a \end{vmatrix} = (a-b)^3 \begin{vmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & a+3b \end{vmatrix} \\ & = (a-b)^3(a+3b) \end{aligned}$$

Question 33. Determine whether the given matrix is invertible by calculating its determinant. If the matrix is invertible, compute its inverse using the adjugate method.

$$1. A = \begin{pmatrix} 5 & 6 \\ -3 & -4 \end{pmatrix}$$

$$2. A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$3. A = \begin{pmatrix} -12 & 7 & -27 \\ 4 & -1 & 2 \\ 3 & 2 & -8 \end{pmatrix}$$

$$4. A = \begin{pmatrix} 31 & -20 & 106 \\ -11 & 7 & -37 \\ -9 & 6 & -32 \end{pmatrix}$$

$$5. A = \begin{pmatrix} 2 & 5 & 5 \\ -1 & -1 & 0 \\ 2 & 4 & 3 \end{pmatrix}$$

$$6. A = \begin{pmatrix} 4 & 2 & 8 \\ -2 & 1 & -4 \\ 3 & 1 & 6 \end{pmatrix}$$

$$7. A = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 3 & 2 \\ -2 & 0 & -4 \end{pmatrix}$$

$$8. A = \begin{pmatrix} 1 & 0 & -1 \\ 9 & -1 & 4 \\ 8 & 9 & -1 \end{pmatrix}$$

$$9. A = \begin{pmatrix} 2 & -3 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{pmatrix}$$

$$10. A = \begin{pmatrix} -3 & 0 & 1 \\ 5 & 0 & 6 \\ 8 & 0 & 3 \end{pmatrix}$$

$$11. A = \begin{pmatrix} 2 & 0 & 0 \\ 8 & 1 & 0 \\ -5 & 3 & 6 \end{pmatrix}$$

$$12. A = \begin{pmatrix} \sqrt{2} & -\sqrt{7} & 0 \\ 3\sqrt{2} & -3\sqrt{7} & 0 \\ 5 & -9 & 0 \end{pmatrix}$$

$$13. A = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 3 & 2 \\ -1 & 0 & -4 \end{pmatrix}$$

Solution.

$$1. \det(A) = -2$$

$$\operatorname{adj}(A) = \begin{pmatrix} -4 & -6 \\ 3 & 5 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} 2 & 3 \\ -\frac{3}{2} & -\frac{5}{2} \end{pmatrix}$$

$$2. \det(A) = 1$$

$$\operatorname{adj}(A) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

$$3. \det(A) = -79$$

$$\operatorname{adj}(A) = \begin{pmatrix} 4 & 2 & -13 \\ 38 & 177 & -84 \\ 11 & 45 & -16 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -\frac{4}{79} & -\frac{2}{79} & \frac{13}{79} \\ -\frac{38}{79} & -\frac{177}{79} & \frac{84}{79} \\ -\frac{11}{79} & -\frac{45}{79} & \frac{16}{79} \end{pmatrix}$$

$$4. \det(A) = 0$$

$$5. \det(A) = -1$$

$$\operatorname{adj}(A) = \begin{pmatrix} -3 & 5 & 5 \\ 3 & -4 & -5 \\ -2 & 2 & 3 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} 3 & -5 & -5 \\ -3 & 4 & 5 \\ 2 & -2 & -3 \end{pmatrix}$$

6. $\det(A) = 0$

7. $\det(A) = -6$

$$\operatorname{adj}(A) = \begin{pmatrix} -12 & 0 & -9 \\ -4 & -2 & -4 \\ 6 & 0 & 6 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} 2 & 0 & \frac{3}{2} \\ 2 & 1 & \frac{2}{3} \\ \frac{3}{-1} & \frac{3}{0} & \frac{3}{-1} \end{pmatrix}$$

8. $\det(A) = -124$

$$\operatorname{adj}(A) = \begin{pmatrix} -35 & -9 & -1 \\ 41 & 7 & -13 \\ 89 & -9 & -1 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{35}{124} & \frac{9}{124} & \frac{1}{124} \\ -\frac{41}{124} & -\frac{7}{124} & \frac{13}{124} \\ -\frac{89}{124} & \frac{9}{124} & \frac{1}{124} \end{pmatrix}$$

9. $\det(A) = 4$

$$\operatorname{adj}(A) = \begin{pmatrix} 2 & 6 & 4 \\ 0 & 4 & 6 \\ 0 & 0 & 2 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{3}{2} & 1 \\ 0 & 1 & \frac{3}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

10. $\det(A) = 0$

11. $\det(A) = 12$

$$\operatorname{adj}(A) = \begin{pmatrix} 6 & 0 & 0 \\ -48 & 12 & 0 \\ 29 & -6 & 2 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ -4 & 1 & 0 \\ \frac{29}{12} & -\frac{1}{2} & \frac{1}{6} \end{pmatrix}$$

12. $\det(A) = 0$

13. $\det(A) = -15$

$$\operatorname{adj}(A) = \begin{pmatrix} -12 & 0 & -9 \\ -2 & -5 & -4 \\ 3 & 0 & 6 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{4}{5} & 0 & \frac{3}{5} \\ \frac{2}{15} & \frac{1}{3} & \frac{4}{15} \\ -\frac{1}{5} & 0 & -\frac{2}{5} \end{pmatrix}$$

Question 34. Evaluate the determinant of the given matrix by

- (a) cofactor expansion
- (b) row operations

$$1. A = \begin{pmatrix} 2 & -2 & -2 & -2 \\ -2 & 2 & 3 & 0 \\ -2 & -2 & 2 & 0 \\ 1 & -1 & -3 & -1 \end{pmatrix}$$

$$2. A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -3 & -3 & -1 & -1 \\ -1 & -1 & -3 & 2 \\ -1 & -2 & 2 & 1 \end{pmatrix}$$

$$3. A = \begin{pmatrix} -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & -1 & 1 & 1 & 0 \end{pmatrix}$$

$$4. A = \begin{pmatrix} 1 & 0 & -1 & 0 & -1 \\ -1 & -1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 & 0 \\ -1 & 1 & 1 & -1 & 0 \end{pmatrix}$$

Solution.

1. (a) Cofactor expansion along the fourth column

$$\det(A) = (-2) \times 12 + (-1) \times 8 = -32$$

(b)

$$|A| \xrightarrow[R_1 \leftrightarrow R_4, R_2 \leftrightarrow R_3]{R_2 \rightarrow R_2 + R_1, R_3 \rightarrow R_3 + R_1} \begin{vmatrix} 2 & -2 & -2 & -2 \\ 0 & 0 & 1 & -2 \\ 0 & -4 & 0 & -2 \\ 1 & -1 & -3 & -1 \end{vmatrix} \xrightarrow{R_1 \rightarrow R_1 - 2R_4} \begin{vmatrix} 0 & 0 & 4 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & -4 & 0 & -2 \\ 1 & -1 & -3 & -1 \end{vmatrix} \\ \xrightarrow{R_1 \leftrightarrow R_4, R_2 \leftrightarrow R_3} \begin{vmatrix} 1 & -1 & -3 & -1 \\ 0 & -4 & 0 & -2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 4 & 0 \end{vmatrix} \xrightarrow{R_4 \rightarrow R_4 - 4R_3} \begin{vmatrix} 1 & -1 & -3 & -1 \\ 0 & -4 & 0 & -2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 8 \end{vmatrix} = -32$$

In exam solutions, all row operations used in determinant computations must be explicitly stated. This is omitted in parts 2-4.

2. (a) Cofactor expansion along the first row

$$\det(A) = 1 \times 32 + (-1) \times (-27) = 59$$

(b)

$$|A| = \begin{vmatrix} 1 & -1 & 0 & 0 \\ 0 & -6 & -1 & -1 \\ 0 & -2 & -3 & 2 \\ 0 & -3 & 2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 0 & 0 \\ 0 & -6 & -1 & -1 \\ 0 & 1 & -5 & 1 \\ 0 & -3 & 2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & -31 & 5 \\ 0 & 1 & -5 & 1 \\ 0 & 0 & -13 & 4 \end{vmatrix} = - \begin{vmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -5 & 1 \\ 0 & 0 & -31 & 5 \\ 0 & 0 & -13 & 4 \end{vmatrix} \\ = - \begin{vmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -5 & 1 \\ 0 & 0 & -31 & 5 \\ 0 & 0 & 0 & \frac{59}{31} \end{vmatrix} = 59$$

3. (a) Cofactor expansion along the second row

$$\det(A) = (-1) \times 0 = 0$$

(b)

$$|A| = \begin{vmatrix} -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 0 \end{vmatrix} = - \begin{vmatrix} -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{vmatrix} = 0$$

4. (a) Cofactor expansion along the third row

$$\det(A) = 1 \times 4 + (-1) \times (-1) = 5$$

(b)

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 0 & -1 & 0 & -1 \\ 0 & -1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & -1 & 0 & -1 \\ 0 & -1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & -1 & -1 & -3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & -1 & 0 & -1 \\ 0 & -1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & -1 & -3 \end{vmatrix} \\ &= \begin{vmatrix} 1 & 0 & -1 & 0 & -1 \\ 0 & -1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & -5 \end{vmatrix} = 5 \end{aligned}$$

Question 35. Solve the following systems of equations using (if possible)

- (a) the inverse of the coefficient matrix computed via the adjugate method
 (b) Cramer's rule

1.

$$\begin{aligned} 7x_1 - 2x_2 &= 3 \\ 3x_1 + x_2 &= 5 \end{aligned}$$

2.

$$\begin{aligned} -9x - 4y &= 3 \\ -7x + 5y &= -10 \end{aligned}$$

3.

$$\begin{aligned} 2x + 3y &= 4 \\ 2x + 2y &= 4 \end{aligned}$$

4.

$$\begin{aligned} -10x - 7y &= -12 \\ 12x - 11y &= 5 \end{aligned}$$

5.

$$\begin{aligned} 5x - 5y &= 7 \\ 2x - 3y &= 6 \end{aligned}$$

6.

$$\begin{aligned} -x - 3y &= 4 \\ -8x + 4y &= 3 \end{aligned}$$

7.

$$\begin{aligned} 2x + 5y &= 4 \\ 4x + y &= 3 \end{aligned}$$

8.

$$\begin{aligned} x_1 + x_2 + 2x_3 &= 8 \\ -x_1 - 2x_2 + 3x_3 &= 1 \\ 3x_1 - 7x_2 + 4x_3 &= 10 \end{aligned}$$

9.

$$\begin{aligned} -2x + y - 4z &= -8 \\ -4y + z &= 3 \\ 4x - z &= -8 \end{aligned}$$

10.

$$\begin{aligned} 4x + 5y &= 2 \\ 11x + y + 2z &= 3 \\ x + 5y + 2z &= 1 \end{aligned}$$

11.

$$\begin{aligned} 2x + 3y + 2z &= -2 \\ -x - 3y - 8z &= -2 \\ -3x + 2y - 7z &= 2 \end{aligned}$$

12.

$$\begin{aligned} x - 4y + z &= 6 \\ 4x - y + 2z &= -1 \\ 2x + 2y - 3z &= -20 \end{aligned}$$

13.

$$\begin{aligned} x_1 - 3x_2 + x_3 &= 4 \\ 2x_1 - x_2 &= -2 \\ 4x_1 - 3x_3 &= 0 \end{aligned}$$

14.

$$\begin{aligned} 3x_1 - x_2 + x_3 &= 4 \\ -x_1 + 7x_2 - 3x_3 &= 1 \\ 2x_1 + 6x_2 - x_3 &= 5 \end{aligned}$$

15.

$$\begin{aligned} 2x_1 + 2x_2 + 2x_3 &= 0 \\ -2x_1 + 5x_2 + 2x_3 &= 1 \\ 8x_1 + x_2 + 4x_3 &= -1 \end{aligned}$$

16.

$$\begin{aligned} -x_1 - 4x_2 + 2x_3 + x_4 &= -32 \\ 2x_1 - x_2 + 7x_3 + 9x_4 &= 14 \\ -x_1 + x_2 + 3x_3 + x_4 &= 11 \\ x_1 - 2x_2 + x_3 - 4x_4 &= -4 \end{aligned}$$

17.

$$\begin{aligned} x - y + 2z - w &= -1 \\ 2x + y - 2z - 2w &= -2 \\ -x + 2y - 4z - 2w &= 1 \\ 3x - 3w &= -3 \end{aligned}$$

Solution.

1. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} 7 & -2 \\ 3 & 1 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 1 & 2 \\ -3 & 7 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{1}{13} & \frac{2}{13} \\ -\frac{3}{13} & \frac{7}{13} \end{pmatrix}$$

The solution of the linear system is $(1, 2)$.

- (b)

$$\det(A) = 13, \quad \det(A_1) = 13, \quad \det(A_2) = 26$$

The solution for the linear system is given by

$$\left(\frac{13}{13}, \frac{26}{13} \right) = (1, 2).$$

2. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} -9 & -4 \\ -7 & 5 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 5 & 4 \\ 7 & -9 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -\frac{5}{73} & -\frac{4}{73} \\ \frac{7}{73} & \frac{9}{73} \end{pmatrix}$$

The solution of the linear system is $\left(\frac{25}{73}, -\frac{111}{73} \right)$.

- (b)

$$\det(A) = -73, \quad \det(A_1) = -25, \quad \det(A_2) = 111$$

The solution of the linear system is $\left(\frac{25}{73}, -\frac{111}{73} \right)$.

3. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} 2 & 3 \\ 2 & 2 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 2 & -3 \\ -2 & 2 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -1 & \frac{3}{2} \\ 1 & -1 \end{pmatrix}$$

The solution of the linear system is $(2, 0)$.

- (b)

$$\det(A) = -2, \quad \det(A_1) = -4, \quad \det(A_2) = 0$$

The solution of the linear system is $(2, 0)$.

4. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} -10 & -7 \\ 12 & -11 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} -11 & 7 \\ -12 & -10 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -\frac{11}{194} & \frac{7}{194} \\ -\frac{6}{97} & -\frac{5}{97} \end{pmatrix}$$

The solution of the linear system is $\left(\frac{167}{194}, \frac{47}{97} \right)$.

(b)

$$\det(A) = 194, \quad \det(A_1) = 167, \quad \det(A_2) = 94$$

The solution of the linear system is $\left(\frac{167}{194}, \frac{47}{97}\right)$.

5. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} 5 & -5 \\ 2 & -3 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} -3 & 5 \\ -2 & 5 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{3}{5} & -1 \\ \frac{2}{5} & -1 \end{pmatrix}$$

The solution of the linear system is $\left(-\frac{9}{5}, -\frac{16}{5}\right)$.

(b)

$$\det(A) = -5, \quad \det(A_1) = 9, \quad \det(A_2) = 16.$$

The solution of the linear system is $\left(-\frac{9}{5}, -\frac{16}{5}\right)$.

6. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} -1 & -3 \\ -8 & 4 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 4 & 3 \\ 8 & -1 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -\frac{1}{7} & -\frac{3}{28} \\ \frac{2}{-7} & \frac{1}{28} \end{pmatrix}$$

The solution of the linear system is $\left(-\frac{25}{28}, -\frac{29}{28}\right)$.

(b)

$$\det(A) = -28, \quad \det(A_1) = 25, \quad \det(A_2) = 29.$$

The solution of the linear system is $\left(-\frac{25}{28}, -\frac{29}{28}\right)$.

7. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} 2 & 5 \\ 4 & 1 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 1 & -5 \\ -4 & 2 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -\frac{1}{18} & \frac{5}{18} \\ \frac{2}{9} & -\frac{1}{9} \end{pmatrix}.$$

(b)

$$\det(A) = -18, \quad \det(A_1) = -11, \quad \det(A_2) = -10$$

The solution of the linear system is $\left(\frac{11}{18}, \frac{5}{9}\right)$.

8. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} 1 & 1 & 2 \\ -1 & -2 & 3 \\ 3 & -7 & 4 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 13 & -18 & 7 \\ 13 & -2 & -5 \\ 13 & 10 & -1 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{1}{4} & -\frac{9}{26} & \frac{7}{52} \\ \frac{1}{4} & -\frac{1}{26} & -\frac{5}{52} \\ \frac{1}{4} & \frac{5}{26} & -\frac{1}{52} \end{pmatrix}.$$

(b)

$$\det(A) = 52, \quad \det(A_1) = 156, \quad \det(A_2) = 52, \quad \det(A_3) = 104.$$

The solution of the linear system is $(3, 1, 2)$.

9. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} -2 & 1 & -4 \\ 0 & -4 & 1 \\ 4 & 0 & -1 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 4 & 1 & -15 \\ 4 & 18 & 2 \\ 16 & 4 & 8 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -\frac{1}{17} & -\frac{1}{68} & \frac{15}{68} \\ -\frac{1}{17} & -\frac{9}{34} & -\frac{1}{34} \\ -\frac{4}{17} & -\frac{1}{17} & -\frac{2}{17} \end{pmatrix}.$$

(b)

$$\det(A) = -68, \quad \det(A_1) = 91, \quad \det(A_2) = 6, \quad \det(A_3) = -180$$

The solution of the linear system is $\left(-\frac{91}{68}, -\frac{3}{34}, \frac{45}{17}\right)$.

10. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} 4 & 5 & 0 \\ 11 & 1 & 2 \\ 1 & 5 & 2 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} -8 & -10 & 10 \\ -20 & 8 & -8 \\ 54 & -15 & -51 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{2}{33} & \frac{5}{66} & -\frac{5}{66} \\ \frac{5}{33} & -\frac{2}{33} & \frac{2}{33} \\ -\frac{9}{22} & \frac{5}{44} & \frac{17}{44} \end{pmatrix}.$$

(b)

$$\det(A) = -132, \quad \det(A_1) = -36, \quad \det(A_2) = -24, \quad \det(A_3) = 12.$$

The solution of the linear system is $\left(\frac{3}{11}, \frac{2}{11}, -\frac{1}{11}\right)$.

11. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} 2 & 3 & 2 \\ -1 & -3 & -8 \\ -3 & 2 & -7 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 37 & 25 & -18 \\ 17 & -8 & 14 \\ -11 & -13 & -3 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{37}{103} & \frac{25}{103} & -\frac{18}{103} \\ \frac{17}{103} & -\frac{8}{103} & \frac{14}{103} \\ -\frac{11}{103} & -\frac{13}{103} & -\frac{3}{103} \end{pmatrix}.$$

(b)

$$\det(A) = 103, \quad \det(A_1) = -160, \quad \det(A_2) = 10, \quad \det(A_3) = 42.$$

The solution of the linear system is $\left(-\frac{160}{103}, \frac{10}{103}, \frac{42}{103}\right)$.

12. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} 1 & -4 & 1 \\ 4 & -1 & 2 \\ 2 & 2 & -3 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} -1 & -10 & -7 \\ 16 & -5 & 2 \\ 10 & -10 & 15 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{1}{55} & \frac{2}{11} & \frac{7}{55} \\ -\frac{16}{55} & \frac{1}{11} & -\frac{2}{55} \\ -\frac{2}{11} & \frac{2}{11} & -\frac{3}{11} \end{pmatrix}.$$

(b)

$$\det(A) = -55, \quad \det(A_1) = 144, \quad \det(A_2) = 61, \quad \det(A_3) = -230,$$

The solution of the linear system is $\left(-\frac{144}{55}, -\frac{61}{55}, \frac{46}{11}\right)$.

13. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} 1 & -3 & 1 \\ 2 & -1 & 0 \\ 4 & 0 & -3 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 3 & -9 & 1 \\ 6 & -7 & 2 \\ 4 & -12 & 5 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -\frac{3}{11} & \frac{9}{11} & -\frac{1}{11} \\ -\frac{6}{11} & \frac{7}{11} & -\frac{2}{11} \\ -\frac{4}{11} & \frac{12}{11} & -\frac{5}{11} \end{pmatrix}.$$

(b)

$$\det(A) = -11, \quad \det(A_1) = 30, \quad \det(A_2) = 38, \quad \det(A_3) = 40$$

The solution of the linear system is $\left(-\frac{30}{11}, -\frac{38}{11}, -\frac{40}{11}\right)$.

14. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 7 & -3 \\ 2 & 6 & -1 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 11 & 5 & -4 \\ -7 & -5 & 8 \\ -20 & -20 & 20 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} \frac{11}{20} & \frac{1}{4} & -\frac{1}{5} \\ -\frac{7}{20} & -\frac{1}{4} & \frac{2}{5} \\ -1 & -1 & 1 \end{pmatrix}.$$

(b)

$$\det(A) = 20, \quad \det(A_1) = 29, \quad \det(A_2) = 7, \quad \det(A_3) = 0.$$

The solution of the linear system is $\left(\frac{29}{20}, \frac{7}{20}, 0\right)$.

15. The coefficient matrix is

$$A = \begin{pmatrix} 2 & 2 & 2 \\ -2 & 5 & 2 \\ 8 & 1 & 4 \end{pmatrix}, \quad \det(A) = 0.$$

16. (a) The coefficient matrix, its adjugate, and inverse are

$$A = \begin{pmatrix} -1 & -4 & 2 & 1 \\ 2 & -1 & 7 & 9 \\ -1 & 1 & 3 & 1 \\ 1 & -2 & 1 & -4 \end{pmatrix}, \quad \text{adj}(A) = \begin{pmatrix} 90 & -63 & 117 & -90 \\ 83 & 3 & -66 & 11 \\ 8 & -15 & -93 & -55 \\ -17 & -21 & 39 & 64 \end{pmatrix},$$

$$A^{-1} = \begin{pmatrix} -\frac{10}{47} & \frac{7}{47} & -\frac{13}{47} & \frac{10}{47} \\ -\frac{83}{423} & -\frac{1}{141} & \frac{22}{141} & -\frac{11}{423} \\ -\frac{8}{423} & \frac{5}{141} & \frac{31}{141} & \frac{55}{423} \\ \frac{17}{423} & \frac{7}{141} & -\frac{13}{141} & -\frac{64}{423} \end{pmatrix}.$$

(b)

$$\det(A) = -423, \quad \det(A_1) = -2115, \quad \det(A_2) = -3384, \\ \det(A_3) = -1269, \quad \det(A_4) = 423.$$

The solution of the linear system is $(5, 8, 3, -1)$.

17. The coefficient matrix is

$$A = \begin{pmatrix} 1 & -1 & 2 & -1 \\ 2 & 1 & -2 & -2 \\ -1 & 2 & -4 & -2 \\ 3 & 0 & 0 & -3 \end{pmatrix}, \quad \det(A) = 0.$$